

Percolation theory on networks

Reference:

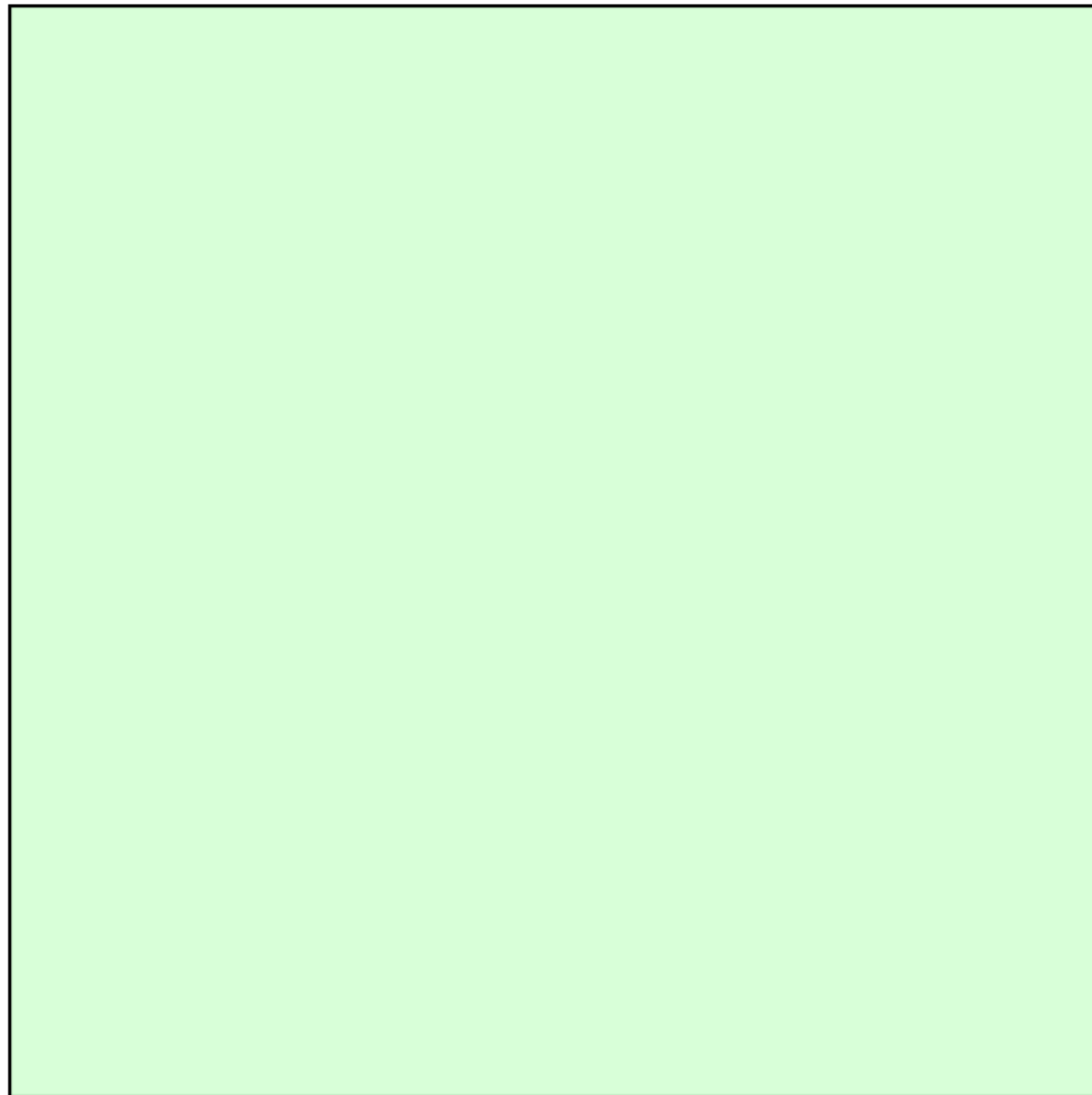
Chapter 15 and 16, Networks, by Mark Newman

Definition

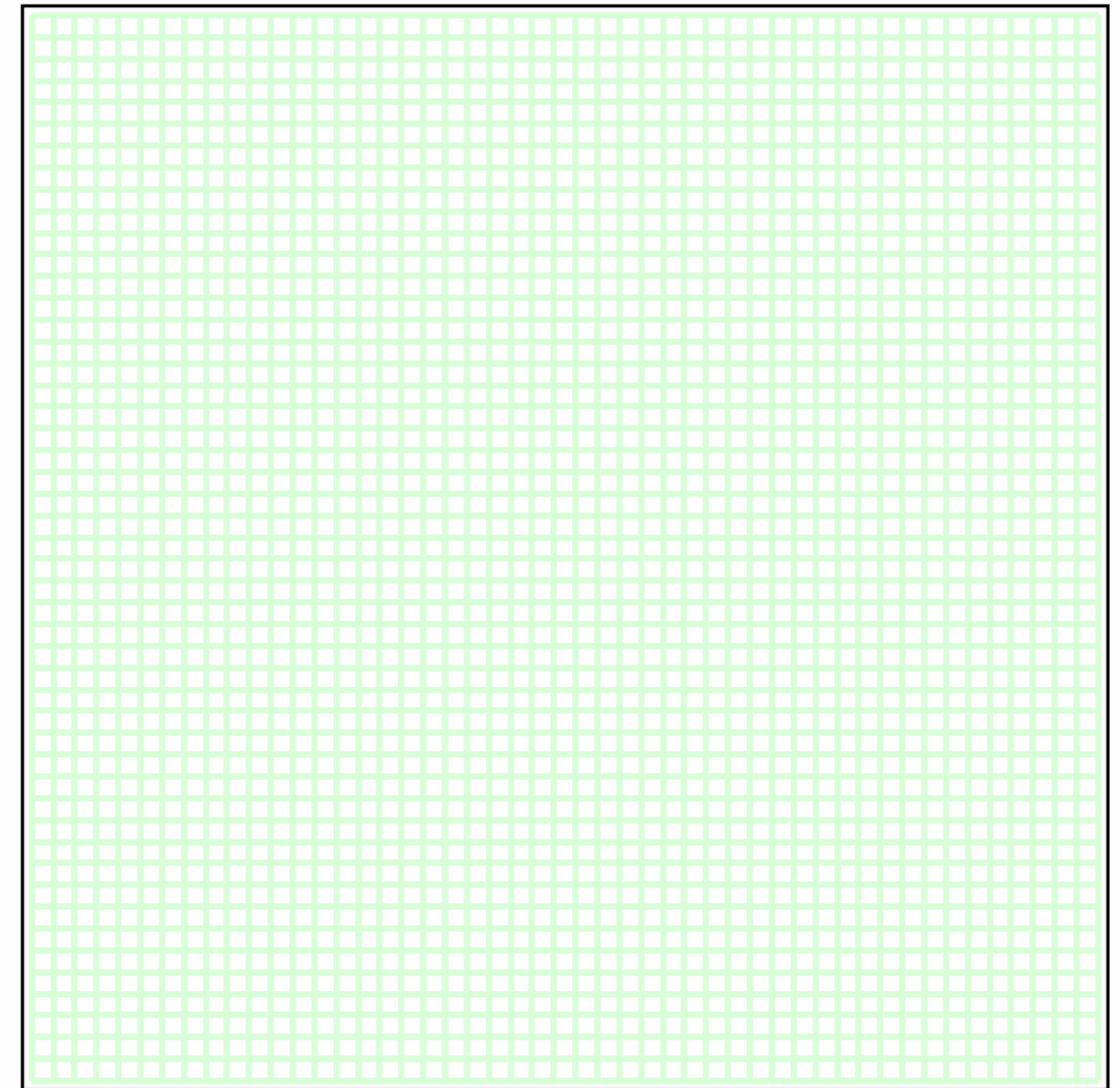
Percolate: trickle through

In material science: “movement of fluid through a porous material”

Site Percolation ($p=0.00$)
Spanning: False



Bond Percolation ($p=0.00$)
Spanning: False



The emergence of a connected giant cluster when the nodes/links are occupied

(Note that we call it ‘giant *cluster*’ not giant *component*, as to distinguish from the previous giant component definition.)

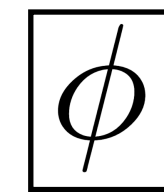
Node (site) percolation - network resilience

Removal of nodes from a network, and study the giant cluster behaviour.

Systems	Nodes (removal mechanism)	Giant cluster	Resilience problems
Internet	Routers (DDOS)	Connected routers	(Partial) loss of internet connectivity
Flight network	Airports (Storm)	Connected airports	Mass cancellation of flights and low connectivity
Train network	Train stations (Strike)	Connected train routes	Similar to airport network
Ecological network	Species (extinction events)	Connected surviving species	Breakdown of food web resulting in mass extinction
Human contact network	Individuals (Vaccination)	Infected cluster	Prevention the mass transmission of infections diseases

Critical transition of (node) percolation

Definitions



: Fraction of nodes NOT removed in the network

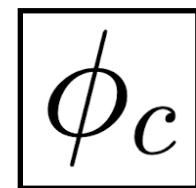
$\phi = 1$ means the original network without node removal

Giant cluster : Connected component that is finite fraction of the whole network
(After nodes removal)

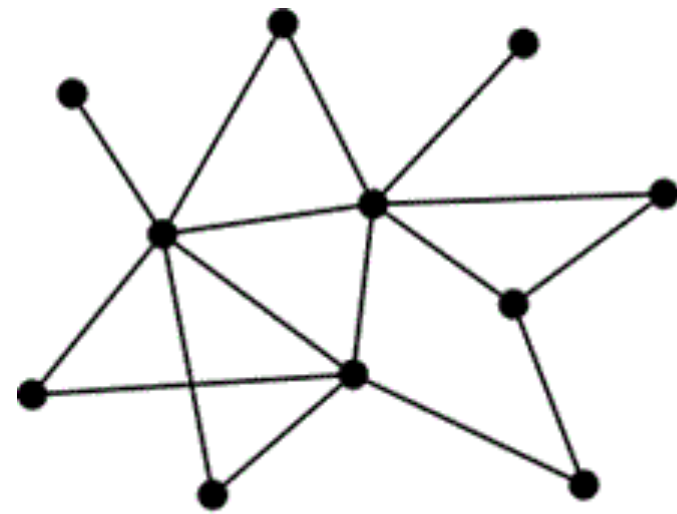
Giant cluster has the same mathematical meaning as giant component

Percolation

: The state when a giant cluster is present



: Percolation threshold - the point when GC start to emerge



(a) $\phi = 1$



(b) $\phi = 0.7$

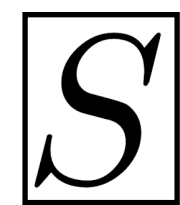


(c) $\phi = 0.3$

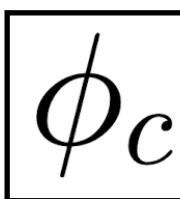


(d) $\phi = 0$

Quantities of interest:



Giant cluster size: how much of the network are still connected (functioning)



Percolation threshold: when it will stop function (no GC)

Random removal of nodes

Probability that a random node is removed: $1 - \phi$

Probability that the node connected through a random link is removed: $1 - \phi$

Probability that a link does not leads to GC: u

Probability that the destination node is not removed, but none of its other k links leads to GC: ϕu^k

If the link leads to a node with excess degree k , probability that it does not lead to GC: $1 - \phi + \phi u^k$

u is also the sum over all possible k excess degrees:

$g_1(z) = \sum_{k=0}^{\infty} q_k z^k$ is the GF of excess degrees

$$u = \sum_{k=0}^{\infty} q_k (1 - \phi + \phi u^k) = 1 - \phi + \phi \sum_{k=0}^{\infty} q_k u^k = 1 - \phi + \phi g_1(u),$$

Probability that a random node (if not removed) of degree k is not connected to GC:

$$\sum_k p_k u^k = g_0(u)$$

$g_0(z) = \sum_{k=0}^{\infty} p_k z^k$ is the GF of original network

Probability that a random node is in GC
- It needs to be not removed AND in GC:

$$S = \phi [1 - g_0(u)]$$

$1 - g_0(u)$ is the probability disregarding removal

This is also the size of GC

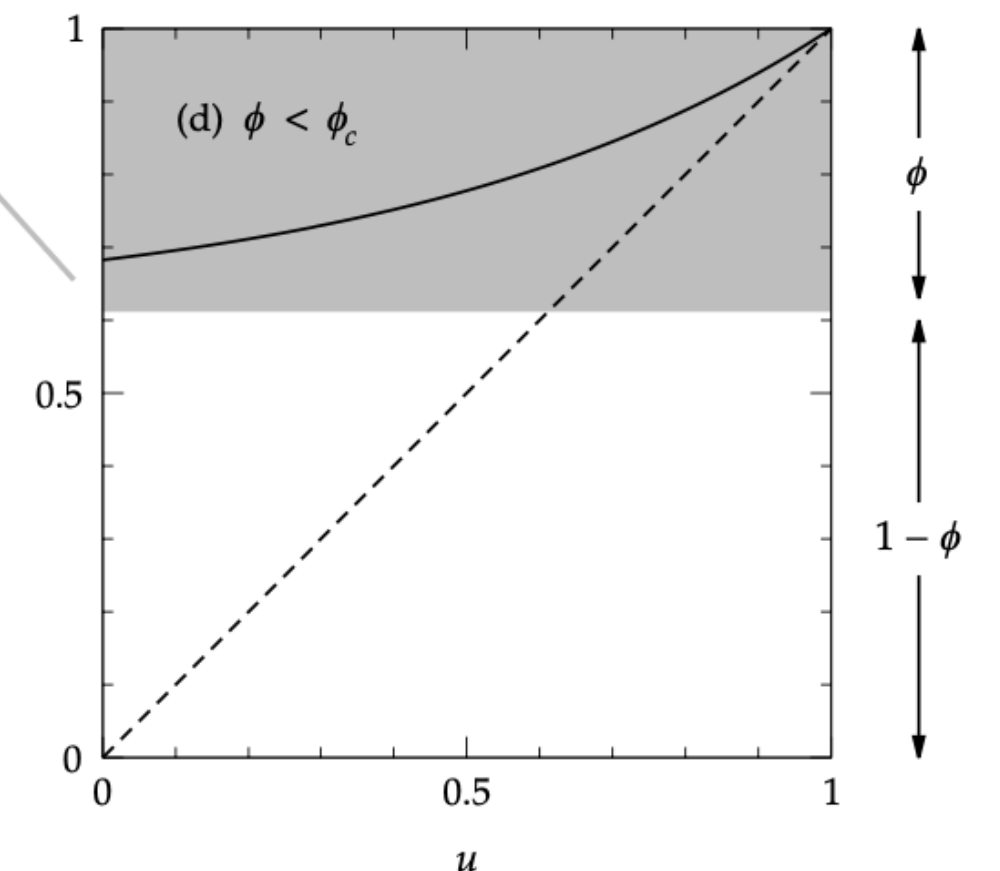
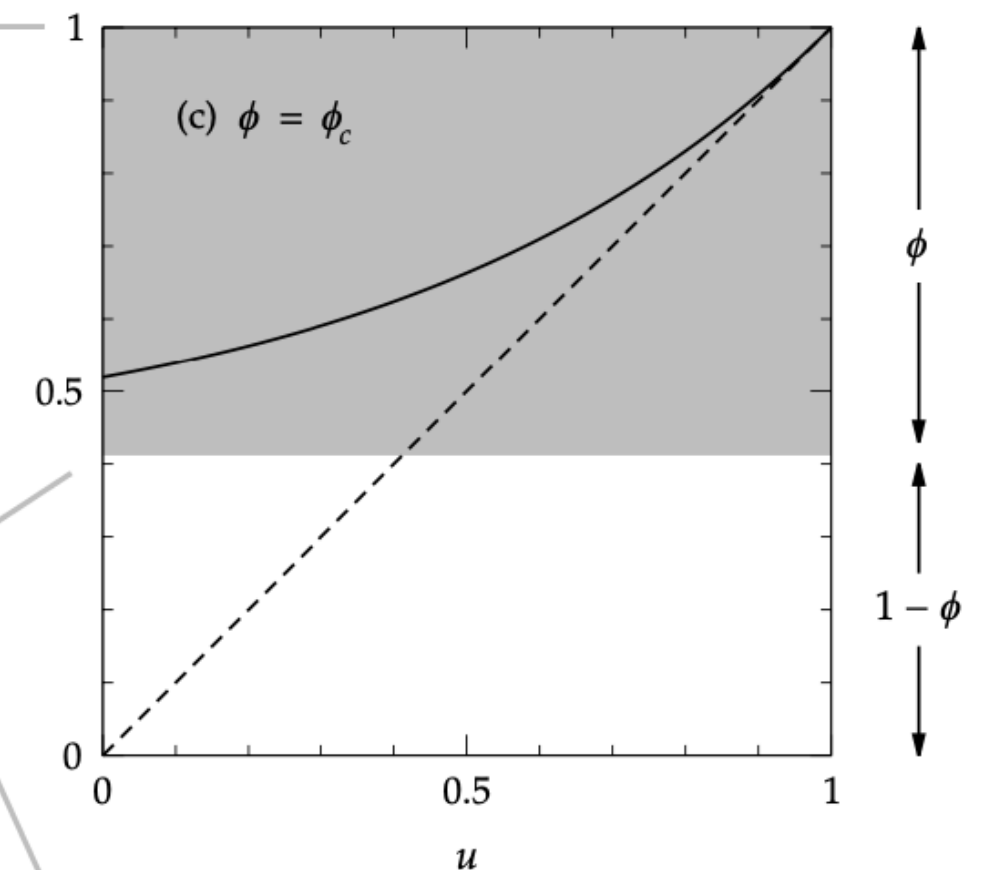
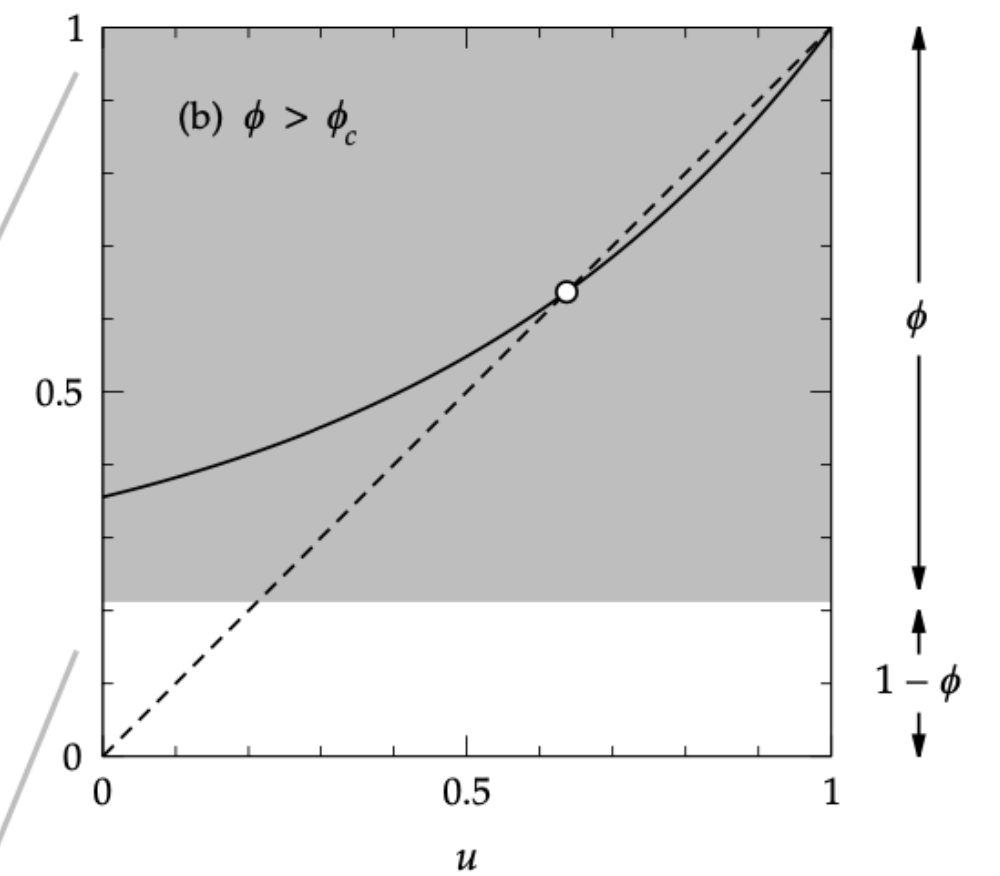
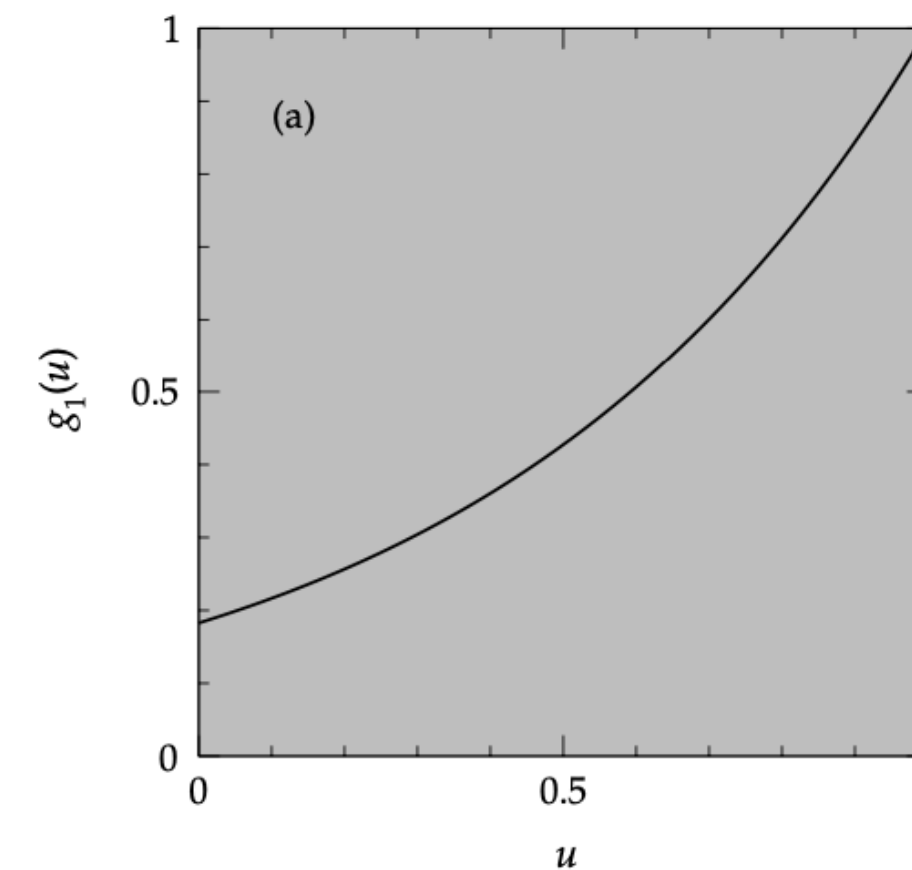
Graphical solution of u :

$$u = 1 - \phi + \phi g_1(u)$$

Find intercept between: $y = u$ **and** $y = 1 - \phi + \phi g_1(u)$

$g_1(1) = 1 \longrightarrow u = 1$ is always a solution

There is always an intercept at $(u, y) = (1, 1)$



**Again, the solution of $u < 1$ appears when GC emerges.
And this happens when two lines touches tangentially at $(1, 1)$**

$$\left[\frac{d}{du} (1 - \phi + \phi g_1(u)) \right]_{u=1} = 1 \longrightarrow \phi_c = \frac{1}{g'_1(1)}$$

Percolation threshold

$$\phi_c = \frac{1}{g'_1(1)} \longrightarrow \phi_c = \frac{\langle k \rangle}{\langle k^2 \rangle - \langle k \rangle} = \frac{1}{\langle k \rangle_{\text{excess}}}$$

Scale free networks are extremely resilient against random node removal

Recall previously the threshold of GC formation without node removal: $\langle k \rangle_{\text{excess}} = \frac{\langle k^2 \rangle - \langle k \rangle}{\langle k \rangle} = 1$

Essentially this means the percolation threshold is simply the inverse of the average excess degree!!!

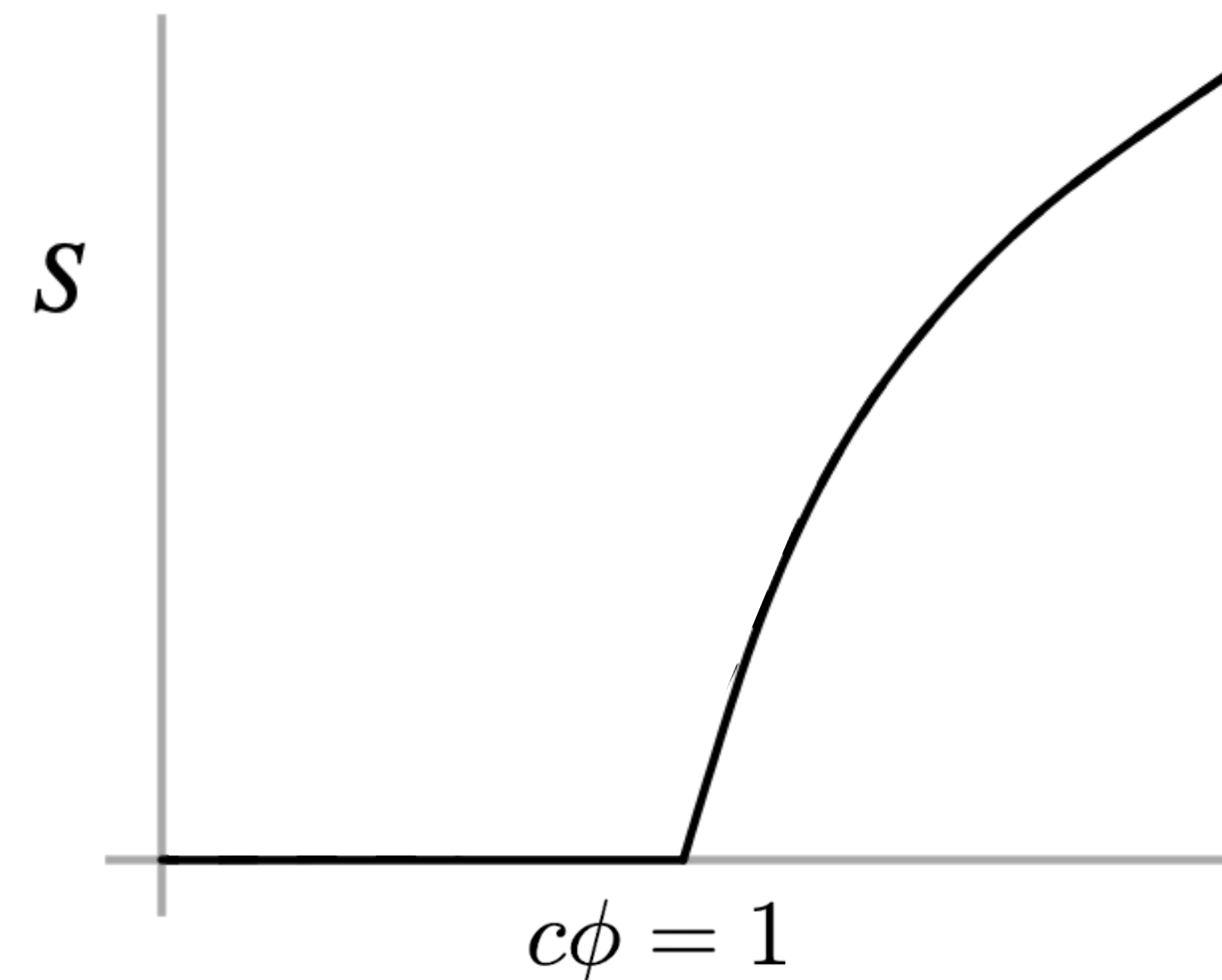
Intuitively, this simply means the ‘effective’ average excess degree = 1 is still the condition for GC formation.

Example with Erdos Renyi network

$$p_k = e^{-c} \frac{c^k}{k!}$$

$$\langle k \rangle_{\text{excess}} = \langle k \rangle = c$$

$$\phi_c = \frac{1}{c}$$



Slope at phase transition

GC size S transitions continuously at the critical point

At critical point, we have: $\boxed{u = 1}$ $\boxed{\phi_c = \frac{1}{g'_1(1)}}$

Expanding $g_1(u)$ around the critical point ϕ_c :

$$\begin{aligned} g_1(u) &= g_1(1) + (u-1)g'_1(1) + \frac{1}{2}(u-1)^2 g''_1(1) + O(u-1)^3 \\ &= 1 + \frac{u-1}{\phi_c} + \frac{1}{2}(u-1)^2 g''_1(1) + O(u-1)^3, \end{aligned}$$

$$u = 1 - \phi + \phi g_1(u)$$

$$u = 1 + \frac{\phi}{\phi_c}(u-1) + \frac{1}{2}\phi(u-1)^2 g''_1(1) + O(u-1)^3$$

$$u - 1 = \frac{2}{g''_1(1)} \frac{\phi_c - \phi}{\phi_c \phi} + O(u-1)^2$$

Expanding $g_0(u)$ around the critical point:

$$g_0(1) = 1 \quad g'_0(1) = \langle k \rangle$$

$$\begin{aligned} g_0(u) &= g_0(1) + (u-1)g'_0(1) + O(u-1)^2 \\ &= 1 + \frac{2\langle k \rangle}{g''_1(1)} \frac{\phi_c - \phi}{\phi_c \phi} + O(\phi - \phi_c)^2, \end{aligned}$$

$$S = \phi[1 - g_0(u)]$$

$$S = \frac{2\langle k \rangle}{g''_1(1)} \frac{\phi - \phi_c}{\phi_c} + \cancel{O(\phi - \phi_c)^2}$$

S increases with $(\phi - \phi_c)$ proportionally near the critical point

Critical exponents

S increases with $(\phi - \phi_c)$ proportionally near the critical point



S is also called the order parameter

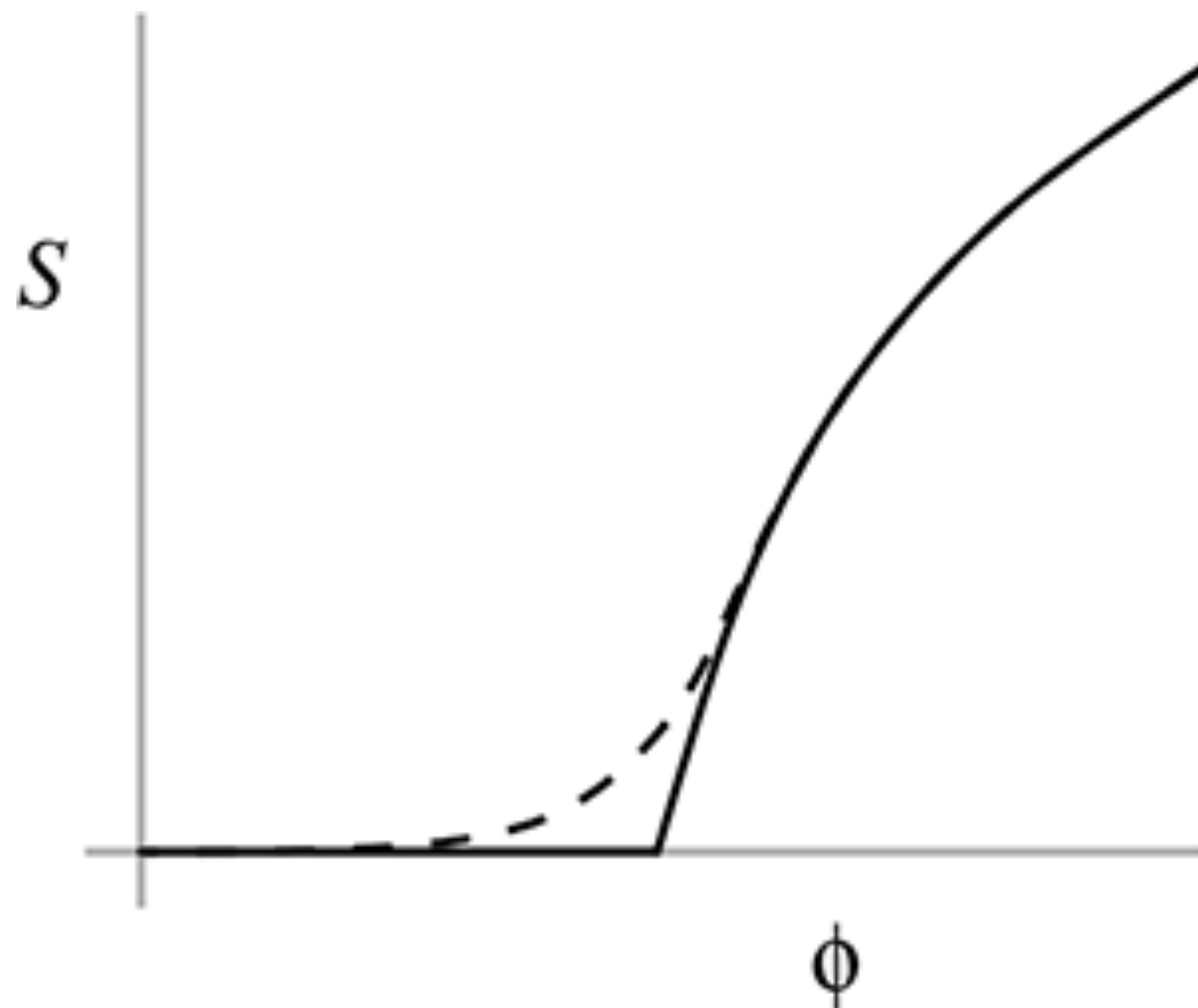


GC size is a continuous phase transition at ϕ_c

Also known as 2_{nd} order phase transition: 2_{nd} derivative is ∞

$$S \sim (\phi - \phi_c)^\beta \quad \beta = 1$$

It turns out, this is one of the several power-laws exponents, also called critical exponents near phase transition:



$$\begin{aligned} S &\sim (\phi - \phi_c)^\beta \\ p_s &\sim s^{-\tau} e^{-|\phi - \phi_c|^{1/\sigma} s} \\ \langle s \rangle &\sim (\phi - \phi_c)^{-\gamma} \\ \xi &\sim |\phi - \phi_c|^{-\nu} \end{aligned}$$

$$\begin{aligned} &\text{General} \\ &\text{scaling} \\ &\text{relations} \\ \beta &= \frac{\tau - 2}{3 - \tau} \\ \gamma &= \frac{\sigma}{\sigma} \end{aligned}$$

$$\begin{aligned} &\text{Random} \\ &\text{networks} \\ \beta_\infty &= 1 \\ \tau_\infty &= 2.5 \\ \sigma_\infty &= 0.5 \\ \gamma_\infty &= 1 \\ \nu_\infty &= 0.5 \end{aligned}$$

Real networks are finite in size, therefore phase transition is not as sharp.

Recall that we have made 'mean-field' assumption, meaning the results is the 'infinite' dimension result.